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Exercise No: 4

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MAP CREATION AND LAYOUT DESIGN USING QGIS

Objective: To import a shapefile in QGIS, create a map layout, and export the final map with coordinates, legend, scale, and north arrow.

Software Required:

- QGIS (latest version installed)
- Shapefile of the study area (water body)

Procedure:

1. Import Shapefile

Open Data Source Manager → Vector → Browse → Select shapefile location → OK.

2. Create Print Layout

Go to Project → New Print Layout → Enter Layout Name → OK.

A new print layout window opens.

3. Add Map to Layout

Go to Add Item → Add Shape → Rectangle.

Draw a rectangle by clicking and dragging from one corner.

Go to Add Item → Add Map → Drag over the rectangle.

Adjust size and position of the shapefile inside the rectangle.

4. Add Coordinates (Grid)

Select the map → Right side panel → Grids → Add a new grid.

Modify Grid:

Enable Grid.

Type: Frame and Annotations Only.

CRS: Use Map CRS.

Interval: 0.1 (both X and Y).

Offset: 0.1.

Frame Style: Line Order.

Enable Draw Coordinates.

Change font style and size.

Format: Degrees and Minutes.

5. Add Legend

Go to Add Item → Add Legend → Place on map.

Default name of shapefile appears.

Use the left panel to modify font, style, and layout.

6. Add Scale Bar

Go to Add Item → Add Scale Bar → Place on map (below or side).

Adjust units, style, and length.

7. Add North Arrow

Go to Add Item → Add North Arrow → Place on map.

8. Export Final Map

Go to Layout → Export as Image.

Choose location, enter file name.

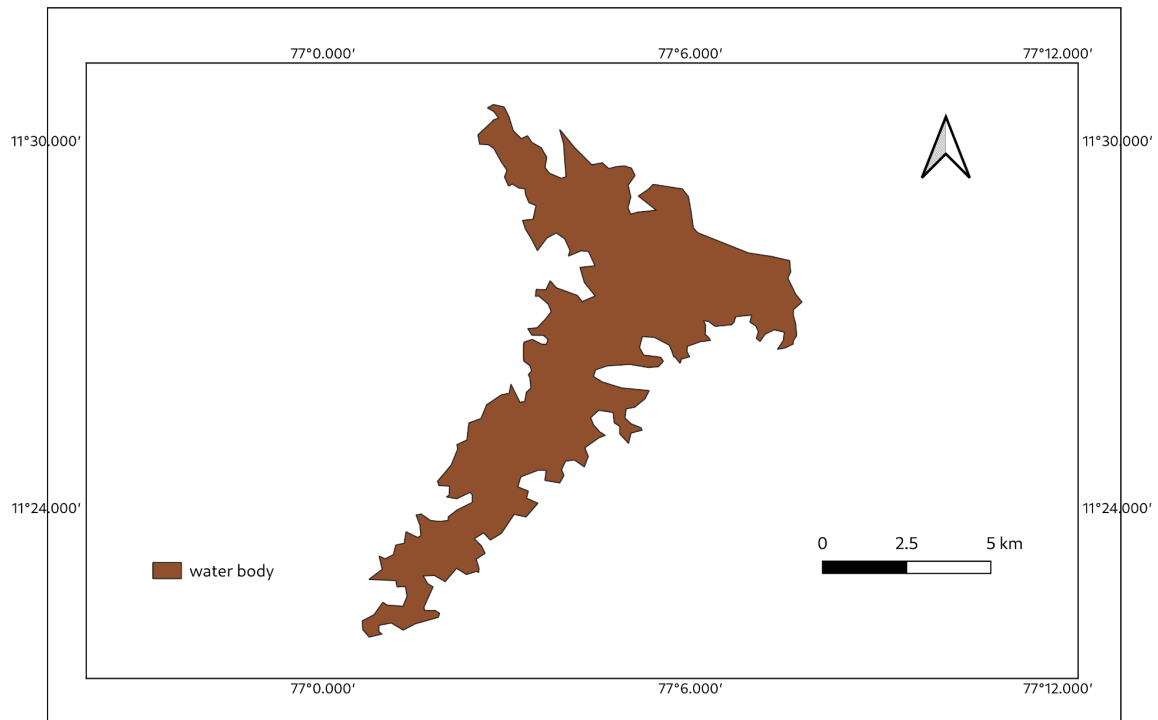
Adjust resolution.

Save as image.

Results:

Map of water body

Image 1



Conclusions:

The exercise demonstrated how to design a printable map in QGIS using the Print Layout tool.

- Imported shapefiles can be styled and placed in layouts.
- Map elements like coordinates, legend, scale bar, and north arrow improve readability.
- The final map can be exported as an image for reports or presentations.

Submitted by

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